

Wenjie Luo

AI Inference and Model Evaluation Engineer | Graduating Jan 2027

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PROFILE

Investigate inference and model behavior with controlled hardware comparisons and numerical gates. Experience in SGLang/vLLM serving, GPU kernels, PyTorch training and per-scene evaluation.

EDUCATION

- National University of Singapore | MSc Civil Engineering**Aug 2025 - Jan 2027
- Transport focus; research in transportation information and control.
- Chang'an University | BEng Transportation Engineering**Sep 2021 - Jun 2025
- GPA 3.16/5.0; approximately top 15% of the major.

EXPERIENCE

- Far East Facade | Project Engineer, Engineering Department**Jun 2026 - Present
- Used Python to extract shipment batches and inspection outcomes from supplier emails, supporting status tracking across 15 façade material types.
- Caixun Technology | Product Intern**Jun 2025 - Aug 2025
- Wrote 3 PRDs with acceptance criteria and delivered 5 Axure prototypes into development; connected requirements to functional validation and defect feedback.

PROJECTS

- SGLang/vLLM Serving and Performance Analysis**Sep 2026
- On a T4 with Qwen2.5-0.5B, traced a SGLang throughput step to default decode-graph capture ending at batch 8; tested the mechanism by disabling graphs and extending capture.
- Setting decode max batch to 32 increased batch-32 throughput from 608 to 2,690 tok/s, or 4.42×, in the same run. Recorded parameters, logs and single measurements per batch.
- fp16x3 Transformer GPU Optimization**Aug 23, 2026 - Present
- Wrote Triton fusion kernels to split operands into fp16 high/low components and feed one expanded-K cuBLAS tensor-core GEMM; combined fused attention and chunking to reduce memory.
- All 13 shapes passed the T4 accuracy gate, with maximum absolute error 9.5e-6. Median speedup was 2.83× across three independent runs per shape; the 100k sequence used 14.2 GB.
- NAVSIM | Planning Model Training and Evaluation**Aug - Sep 2026
- Trained MLP planners on 103,288 navtrain scenes and evaluated 12,146 navtest scenes. Used a map-by-agent 2×2 design with three seeds per cell and paired 95% intervals.
- Reproduced ConstantVelocity 20.7 (published 20.6) and Human 94.6 (94.8). Adding boxes to map-aware models reduced PDMS significantly in all three seeds despite similar L1.

SKILLS

Inference: SGLang, vLLM, CUDA C++, Triton, PyTorch, CUDA Graphs
Evaluation: controlled experiments, accuracy gates, paired bootstrap, factorial designs, multiple seeds